

Simulation of Solar Photolysis of Organic Hydroperoxides: tert-butyl hydroperoxide

Organic hydroperoxides (ROOH) exist as an important atmospheric intermediate, playing a vital role in atmospheric and environmental photochemistry.¹ ROOH are formed via the oxidation and combustion of volatile organic compounds in the atmosphere and the RO-OH bond dissociation from UV light produces reactive OH radicals that enables atmospheric oxidation cycles.²⁻⁵ Previous experimental evidence shows that from UV radiation, the O-O bond of various organic hydroperoxides can dissociate on a repulsive excited state leading to RO+OH radical products.⁴ However, experimentally, it is unclear which electronic state and through which nonadiabatic pathway the RO-OH bond dissociation process takes place. Prior experimental study from measurements of total kinetic energy release (TKER) distribution of the RO+OH show that there are two TKER components, a high- and low-TKER component. The study suggested that each component corresponded to two possible mechanisms, either a direct dissociation on the S_1 state (high-TKER) or an internal conversion process through an S_1/S_0 conical intersection followed by ground-state dissociation (low-TKER)⁴.

Tert-butyl hydroperoxide (TBHP) is a model organic hydroperoxide that, after solar radiation, has excitations into low-lying electronic states to eventually lead towards O-O bond dissociation into hydroxyl and tert-butoxy radical.^{3,4} To determine the mechanistic pathway and electronic state that the O-O bond cleavage occurs on, multireference electronic structure calculations and nonadiabatic molecular dynamics is required to accurately describe the electronic states and bond breaking process. This work uses the state-averaged complete active space self-consistent field (SA-CASSCF) method with *ab initio* multiple spawning (AIMS) nonadiabatic dynamics in the OpenMolcas⁶ and PySPAWN⁷ interface to study the excited-states and photodissociation process of TBHP. This is done through investigating the singlet-spin electronic states of the TBHP potential energy surface (PES) along the O-O bond stretching and simulating the excited-state dynamics to determine which electronic state governs O-O bond dissociation and whether dissociation occurs before or after transfer through a S_1/S_0 conical intersection.

First, an electronic structure validation was completed to obtain a representative active space, optimized geometries, and potential energy surface along the RO-OH bond stretching. The active space is a 6e,4o active space that is composed of O-O σ and σ^* orbitals and two lone pair nonbonding orbitals on the hydroperoxyl oxygen. These 4 orbitals were chosen since they are directly involved in the $n \rightarrow \sigma^*$ transition that governs the S_0/S_1 excitation and O-O bond breaking⁴. The choice of 4 states being averaged was necessary since the S_2 and S_3 states lie close in energy to S_0 and S_1 and the inclusion of more states helps stabilize and balance along the bond-breaking coordinate, even though the entire bond dissociation process is focused on S_0 and S_1 . This active space was generated through a SA-CASSCF calculation averaged over the 4 lowest energy singlets states using the cc-pVDZ⁸ basis set.

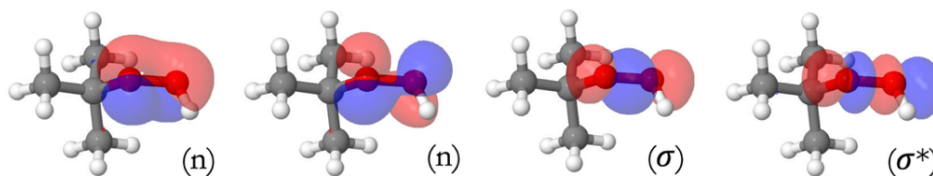


Figure 1. Active space orbitals of TBHP

To investigate the energetics of the bond dissociation process, a relaxed potential energy scan was generated along the O-O bond stretching. The relaxed curves, with respect to S_1 , were calculated at the SA(4)-CASSCF(6e,4o) level of theory. It is evident that the S_1 surface is steep and repulsive across the O-O bond stretching coordinate, indicative of rapid bond dissociation. Moreover, S_1 becomes degenerate with S_0 at large O-O distances.

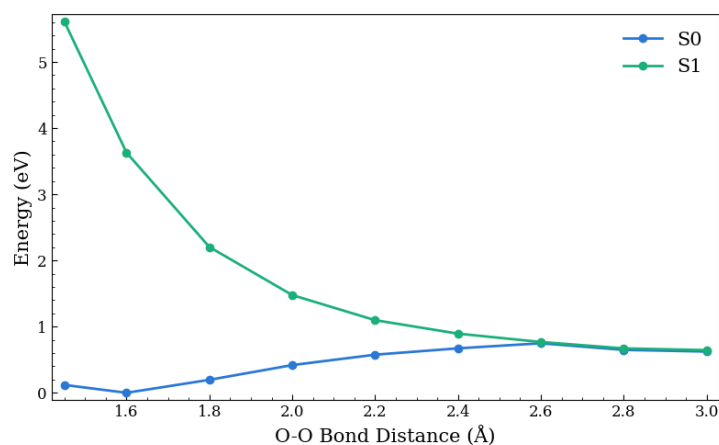


Figure 2. Potential energy scan along the RO-OH bond dissociation pathway for the ground and first excited singlet state. Relaxed curves (with respect to S_1) were computed at the SA(4)-CASSCF(6e,4o) level of theory. The zero of energy was taken relative to the S_0 minimum.

Additionally, dynamics calculations were carried out using PySPAWN interfaced with OpenMolcas. In OpenMolcas, a Hessian calculation of the S_0 equilibrium geometry of TBHP at the SA(4)-CASSCF(6e,4o)/cc-PVDZ level of theory was performed for preparing initial conditions from a Wigner distribution. From this, 50 initial conditions were created, and 50 individual AIMS trajectories were initiated for 750 fs with 0.5 fs time step. The results of the AIMS trajectories reveal that All 50 trajectories have a smooth and continuous RO-OH bond elongation from the Frank-Condon region of $r(\text{O-O}) = 1.4\text{-}1.6 \text{ \AA}$, reaching $2.5\text{-}3.0 \text{ \AA}$ by 750 fs. This is indicative of the bond dissociation process taking place on the repulsive S_1 surface as is shown in figure 2.

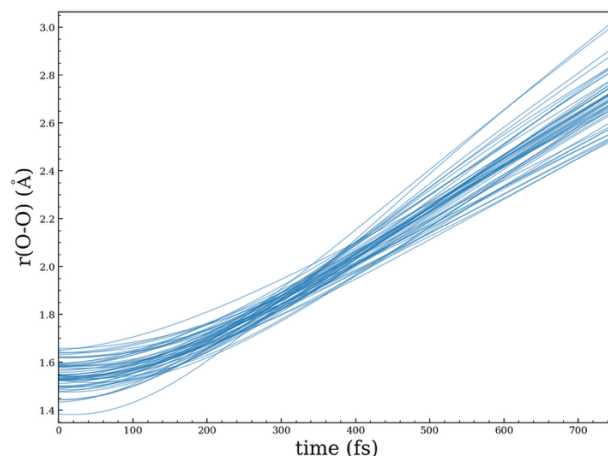


Figure 3. Evolution of the RO-OH bond dissociation as a function of time.

For the analysis of the trajectories, the RO-OH bond cleavage threshold is defined to be 2.6 Å as an operational choice of being ~ 1 Å above the equilibrium geometry RO-OH bond length (1.4-1.6 Å). 2.6 Å is also beyond the steep portion of repulsive S_1 curve and lies at the beginning of the electronic state degeneracy between S_1 and S_0 . With this, the S_1 and S_0 energy gap was tracked along the trajectory progress. It is shown in figure 4 that the energy gap is spread from about 3 to 5 eV for the FC region's vertical excitations at trajectory initiation. As time passes, the energy gap systematically tends towards zero. This is consistent with the behavior seen in figure 2 where the S_1 and S_0 energy curves become degenerate.

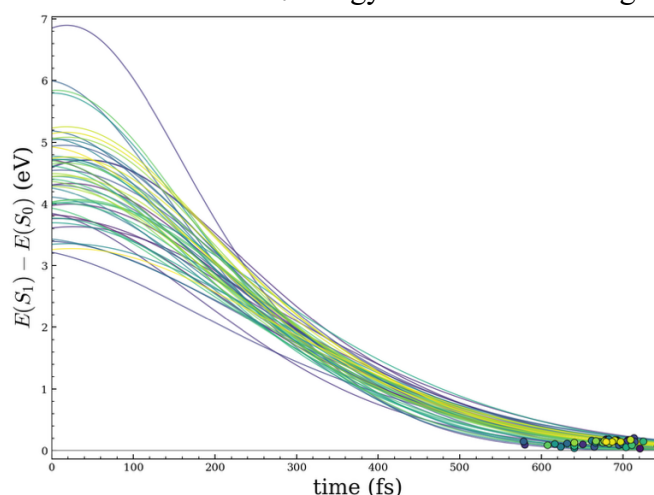


Figure 4. Evolution of the S_1 and S_0 energy gap as a function of time. The time when each trajectory crosses $r(\text{O-O}) = 2.6$ Å is marked with a dot.

It is important to note that throughout the entire progress of the RO-OH bond cleavage, there were no observed spawns to the ground state. All 50 trajectories initiated on S_1 remained on S_1 during the 750 fs simulation. This solidifies that for this 50 trajectory ensemble, the RO-OH bond dissociation occurs on the repulsive S_1 curve rather than through internal conversion and strong coupling to the ground state. Our finding is consistent with the PES of figure 2, which within the range of up to $r(\text{O-O}) = 3$ Å, does not show an obvious conical intersection. In AIMS, spawning is dependent on the nonadiabatic coupling vector or time derivative coupling rather

than just the electronic state energy gap. Due to this, the near degeneracy shown in the PES of figure 2 may not guarantee spawning onto different electronic states.

Collectively, these results and the trajectory ensemble, sampled from the Wigner sampling of ground state TBHP, could indicate a predominant sampling from O-O stretching coordinates. An extended sampling from torsional or O-O-H bending coordinates could provide different results regarding the presence of spawning in the simulations. Ultimately, the lack of observed spawnings to the ground state does not necessarily rule out the mechanistic pathway of internal conversion through an S_1/S_0 conical intersection given the limited initial conditions used in dynamics.

Future work would involve an extended study with cyclohexyl and cyclopentyl hydroperoxide, 2 additional hydroperoxides that have very similar photochemical properties.⁴ Additionally, initiating more than 50 initial conditions for dynamics would better sample the Wigner distribution and provide a more comprehensive representation of the system. Running dynamics with a higher level of theory, such as CASPT2⁹ and PDFT¹⁰ methods would also improve results and recover uncaptured dynamic correlation that is neglected in CASSCF.

References

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Supplemental Figures

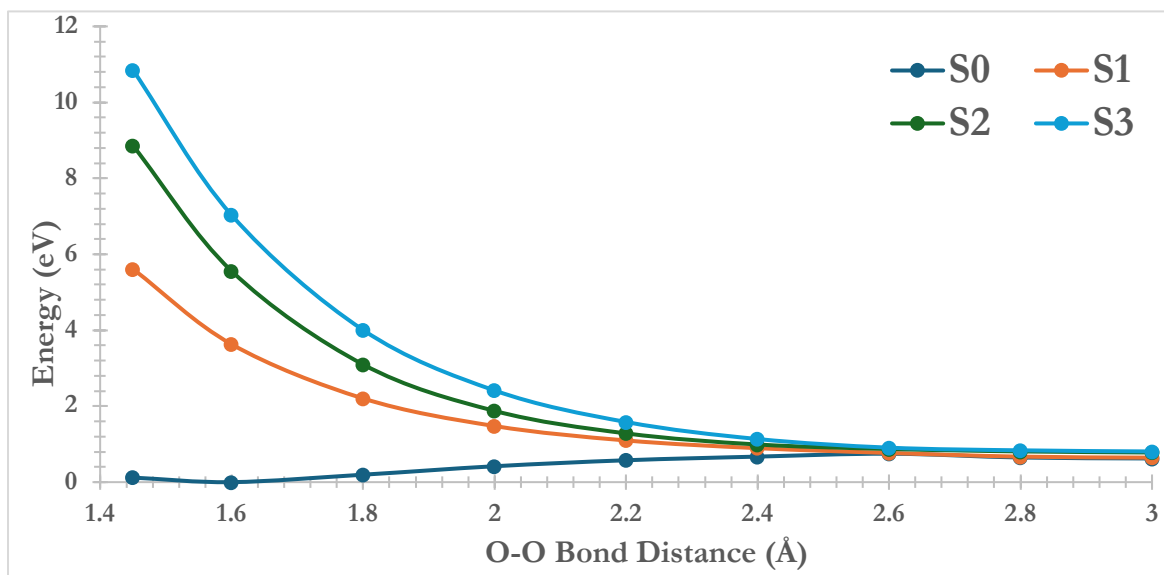


Figure S1: Potential energy scan along the RO-OH bond dissociation pathway for the ground and first three excited singlet states. Relaxed curves (with respect to S₁) were computed at the SA(4)-CASSCF(6e,4o) level of theory. The zero of energy was taken relative to the S₀ minimum.